Master of Computer Applications

Union Christian College Aluva

Software Design Specification

for

Multi-Tenant Manufacturing Management System

Submitted to:

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Roll No & Batch: 18 & 4A

Date of submission:

1. Introduction

1.1 Purpose

The purpose of this Software Design Specification (SDS) is to provide a comprehensive architectural and component-level design for the Multi-Tenant Manufacturing Management System. It serves as the primary reference for developers, testers, and stakeholders, detailing the system's modules, data structures, implementation logic, and user interfaces.

This document aims to ensure:

Traceability: Mapping requirements for specific design components (POS, Stock, Manufacturing).

Scalability: Defining a modular architecture (Microservices-ready, Monorepo) that supports future growth.

Maintainability: establishing clear separation of concerns between Frontend, Backend/API, and Database layers.

1.2 Scope

Multi-Tenant Manufacturing Management System is a cloud-native ERP solution designed specifically for Jewellery Manufacturing and Retailing. The system encompasses the entire detailed lifecycle of jewellery operations:

1. **Master Data Management:** Centralized control of inventory definitions (Diamonds, Stones, Metals), Parties (Customers, Vendors), and Org structure.

2. **Stock Management:** end-to-end tracking of Raw Materials (Gold, Alloys, Loose Stones) from `Stock Entry` (Intake) to `Alloy Creation` (Manufacturing) and Adjustments.
3. **POS & Sales:** A robust Point-of-Sale interface for selling Finished Goods (`Sales Order`), trading items (`Purchase Entry`), and buying scrap (`Old Gold Purchase`).
4. **Reporting:** Real-time visibility into Stock balances, Sales performance, and Inventory movements.

External Interfaces:

Database: PostgreSQL (via Prisma ORM) for persistent storage.

Frontend: Next.js (React) based Web Interface.

Printing: Integration with thermal/laser printers for Invoices and Barcodes.

1.3 Definitions & Acronyms

- **JEMS:** Jewellery Enterprise Management System.
- **POS:** Point of Sale.
- **SDS:** Software Design Specification.
- **UOM:** Unit of Measure (e.g., Grams, Carats).
- **VA:** Value Addition (Making Charges).
- **Old Gold:** Scrap gold jewelry purchased from customers.
- **KYC:** Know Your Customer (Identity Verification).
- **Prisma:** The ORM (Object-Relational Mapping) tool used for database interactions.

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2. Decomposition Description

This section details the functional decomposition of the system into distinct modules as observed in the application architecture.

2.1 Master Data Management

This module is responsible for defining the static data entities used across the system.

Organization Masters:

Tenant: Represents the root organization/company using the SaaS platform.

Department/Role/Employee: Manages internal users, their hierarchy, and access permissions (RBAC).

Inventory Masters:

Metal & Alloy: Defines `MetalPurity` (e.g., 22K), `MetalColour` (Yellow, Rose), and `AlloysRecipe` (Composition of metals).

Diamond Master: Catalog of diamonds defined by Shape, Size (Sieve/mm), Colour, Clarity, and Cut. Linked to `DiamondCertificate` (GIA/IGI). *

Stone Master: Catalog of precious/semi-precious stones defined by Type, Shape, Size, Quality, and Colour.

Product Master: The central catalog of Finished Goods (Jewelry Designs), linking Metals, Diamonds, and Stones into a sellable SKU. Includes Category and Subcategory management.

Party Masters:

Customer: Manages retail client profiles, KYC documentation, and interaction history.

Vendor: Manages suppliers for Raw Materials and Finished Goods.

2.2 Stock Management

This module handles the physical inventory flow of Raw Materials.

Stock Entry: The primary interface for adding inventory. It handles the intake of:

Raw Metals: Pure Gold/Silver bars or grains.

Loose Diamonds/Stones: Packets of stones.

Alloys: Pre-mixed alloy grains.

**Features:* Vendor linkage, Invoice Reference, and Initial Weight capture.

Alloy Creation: A manufacturing process where Raw Metals (Pure Gold) and Alloy Grains are combined (melted) to create a specific Alloy (e.g., 22K Casting Gold).

* *Process:* Inputs -> Wastage Calculation -> Output Stock Lot. *

Stock Adjustment: Allows manual correction of stock levels (Weight/Pieces) due to loss, audit discrepancies, or weighing scale errors.

Stock Report: Real-time ledger of all stock lots, showing `Quantity on Hand`, `Available Qty`, and Purity.

2.3 POS Sales & Trading

This module manages customer-facing transactions.

Sales Order (POS): The core billing interface.

* *Functions:* Customer Selection -> Product Scanning/Lookup -> Discount Application -> Tax Calculation (GST) -> Payment -> Invoice Generation. *

Purchase Entry: Handles the purchase of **Finished Goods** (Trading Stock) from Vendors, distinct from Raw Material Stock Entry.

Old Gold Purchase: A specific workflow for buying scrap jewelry from customers. * *Functions:* Net Weight Calculation (Gross - Stone - Dirt) -> Purity Testing -> Valuation (based on Daily Rate) -> Payment/Voucher Issuance.

POS Stock Management:

POS Stock Report: Tracks inventory specifically allocated to the Showroom/POS floor.

POS Adjustments: Adjustments specific to showroom inventory.

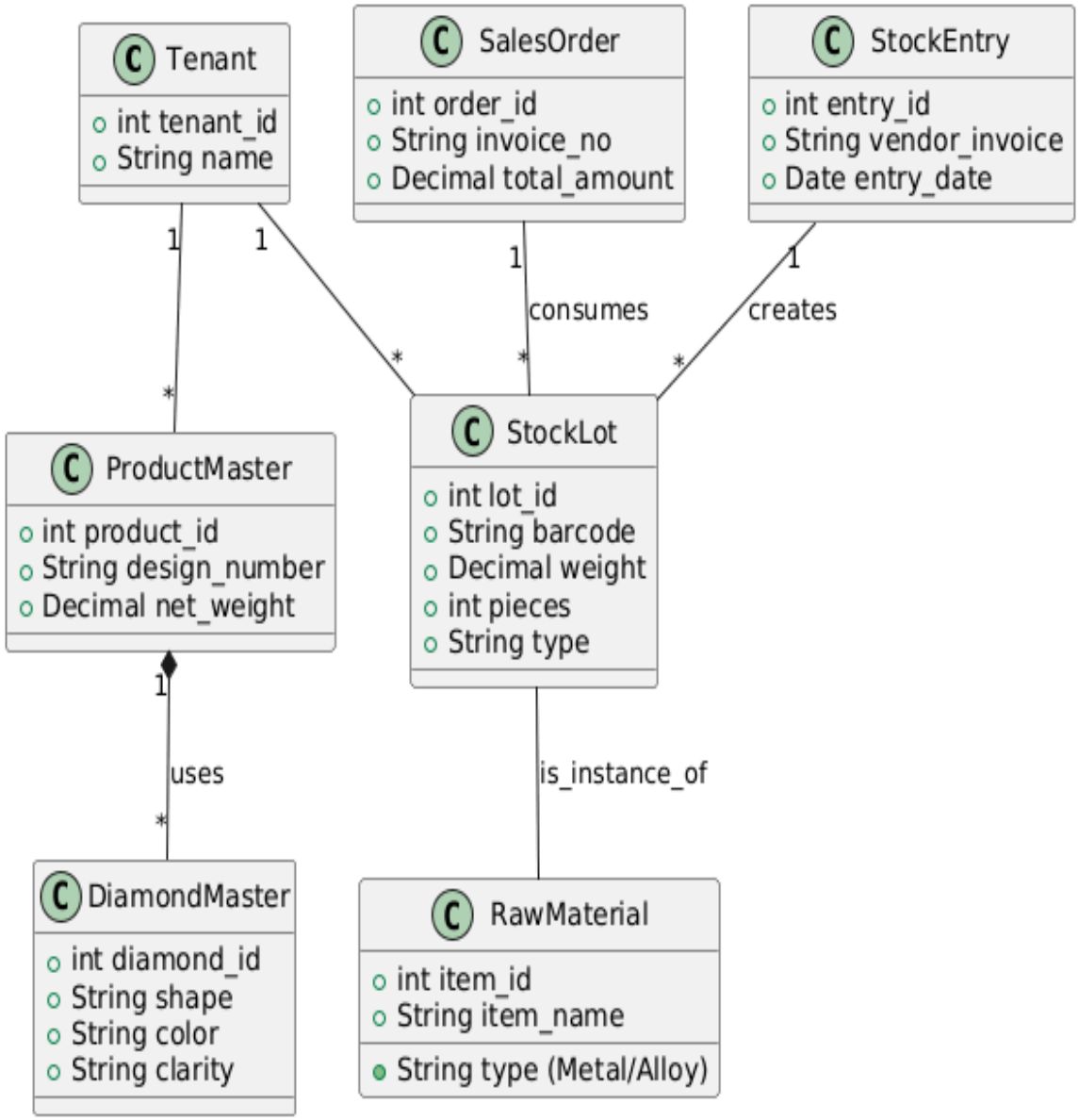
3.4 2.4 Reports

- **Sales Report:** Detailed analysis of Sales Orders by Date, Salesman, Customer, or Product Category.
- **Order Report:** Tracking of custom Job Orders.

4 3. Domain Model

(Analysis Level Class Diagram)

This diagram represents the conceptual entities and their high-level relationships.

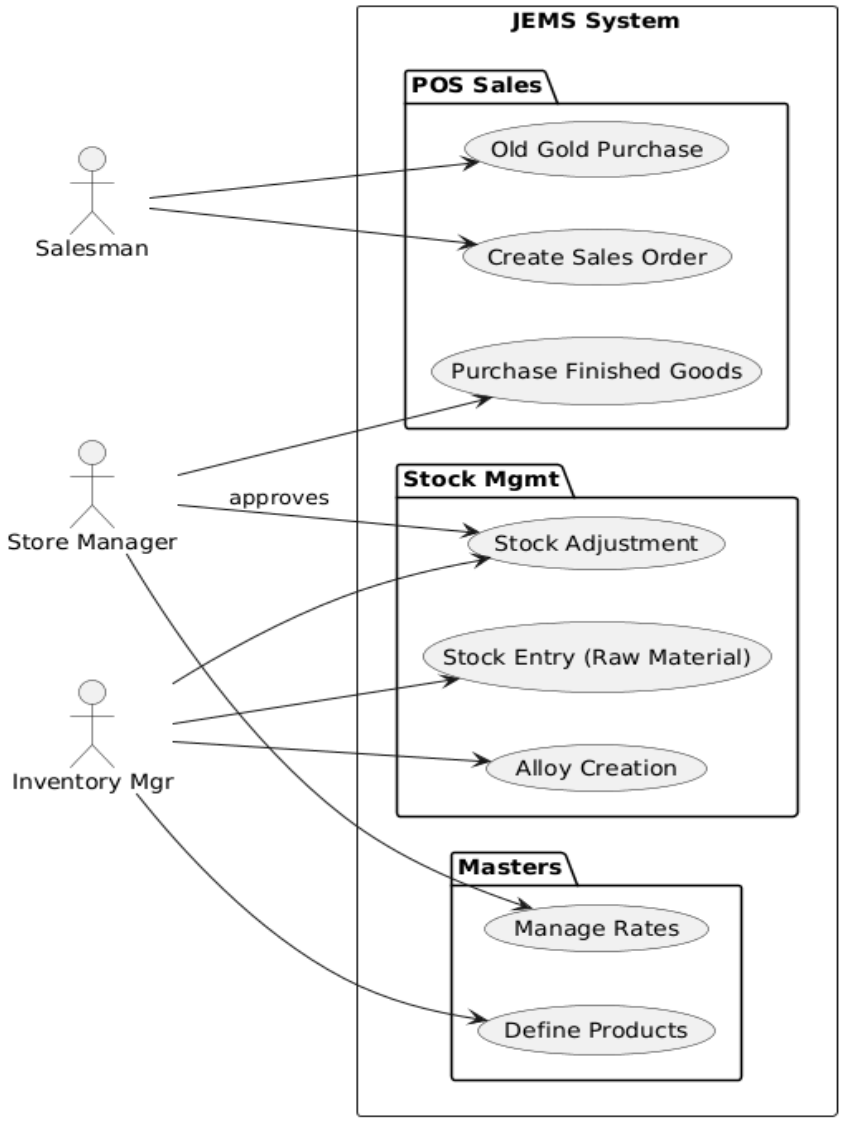


4. Design Model

4.1 Architectural Diagrams

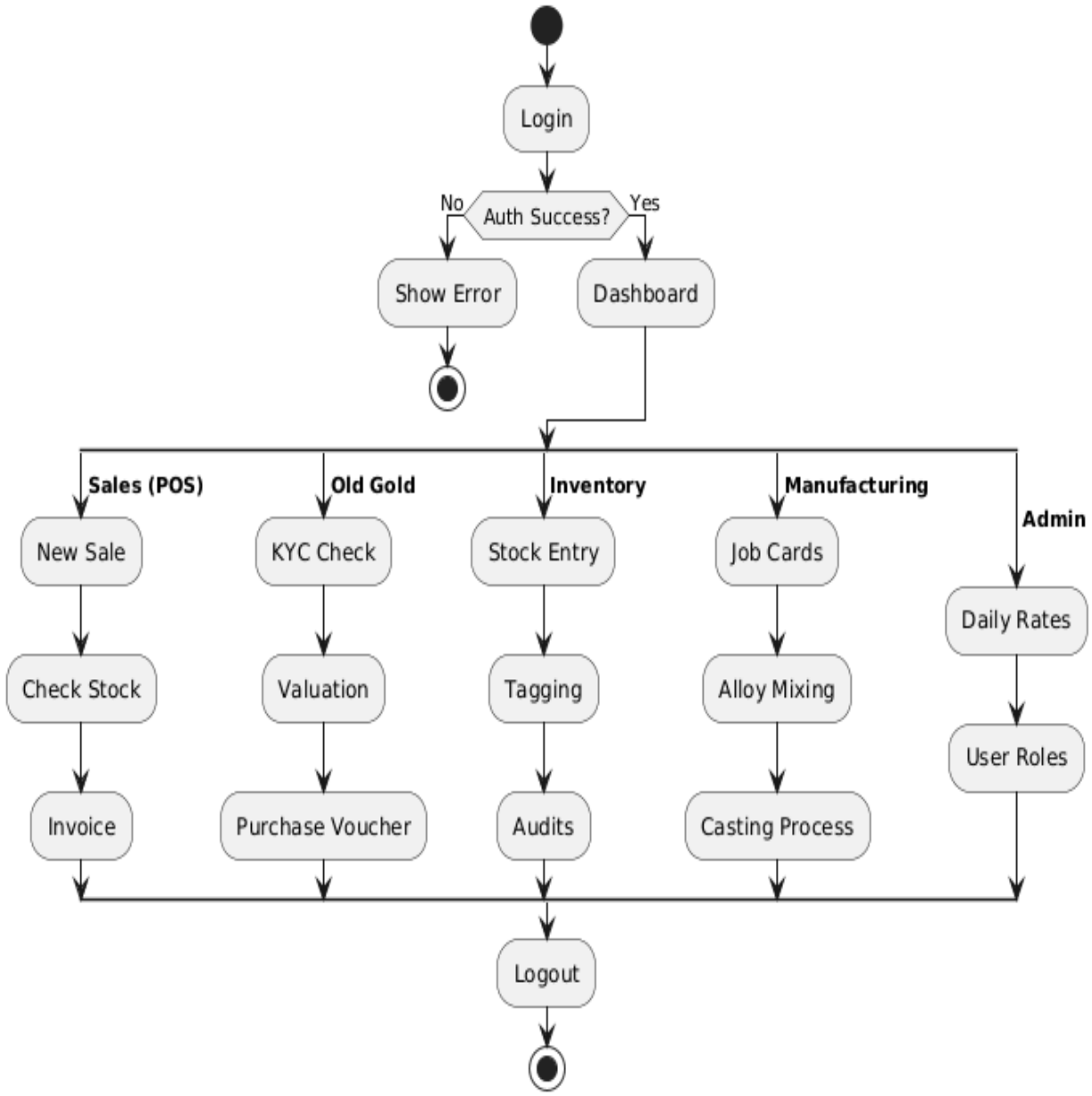
4.1.1 Use Case Diagrams

System-Wide Use Cases

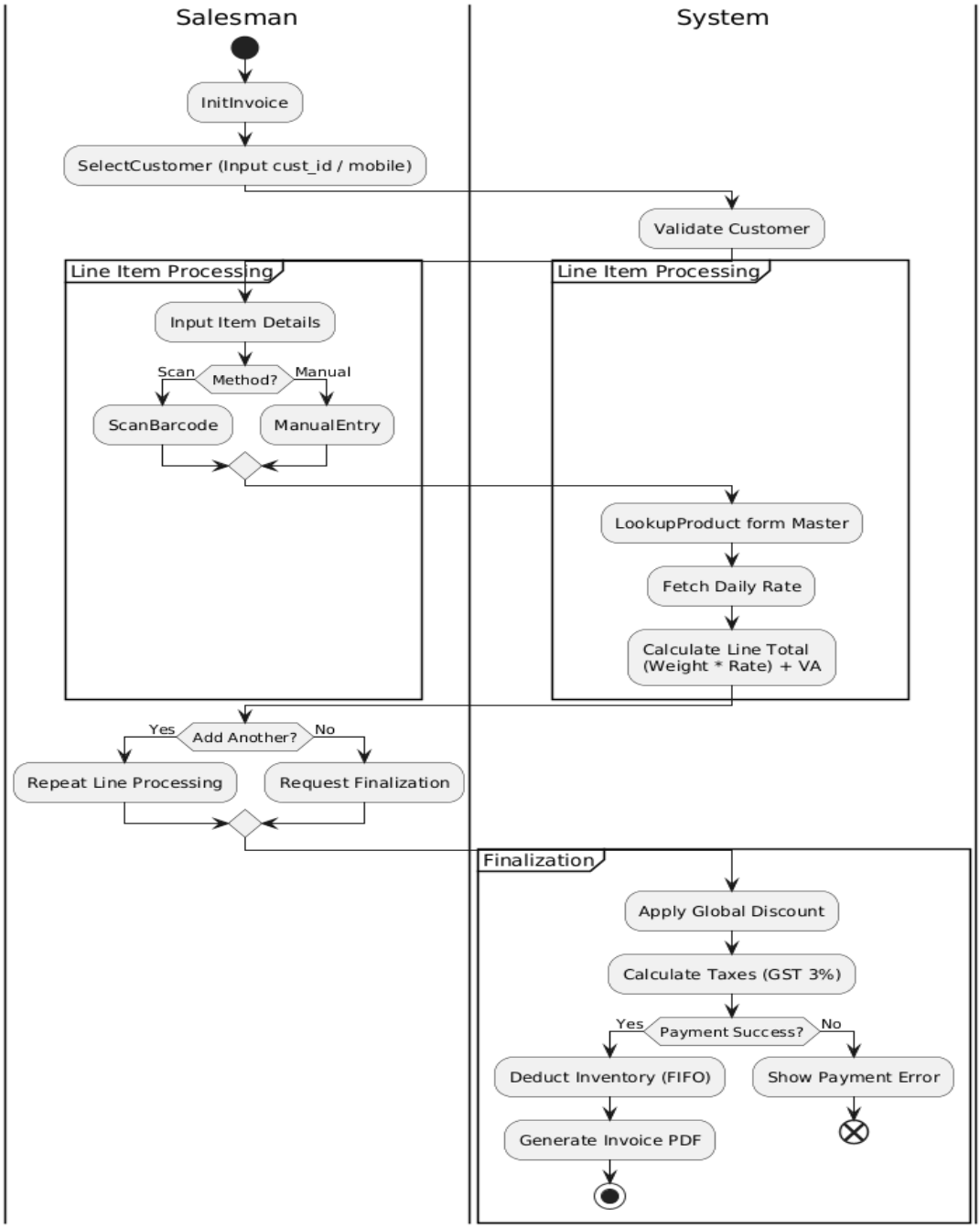


4.1.2 Activity Diagrams

Activity 1: High-Level System Flow

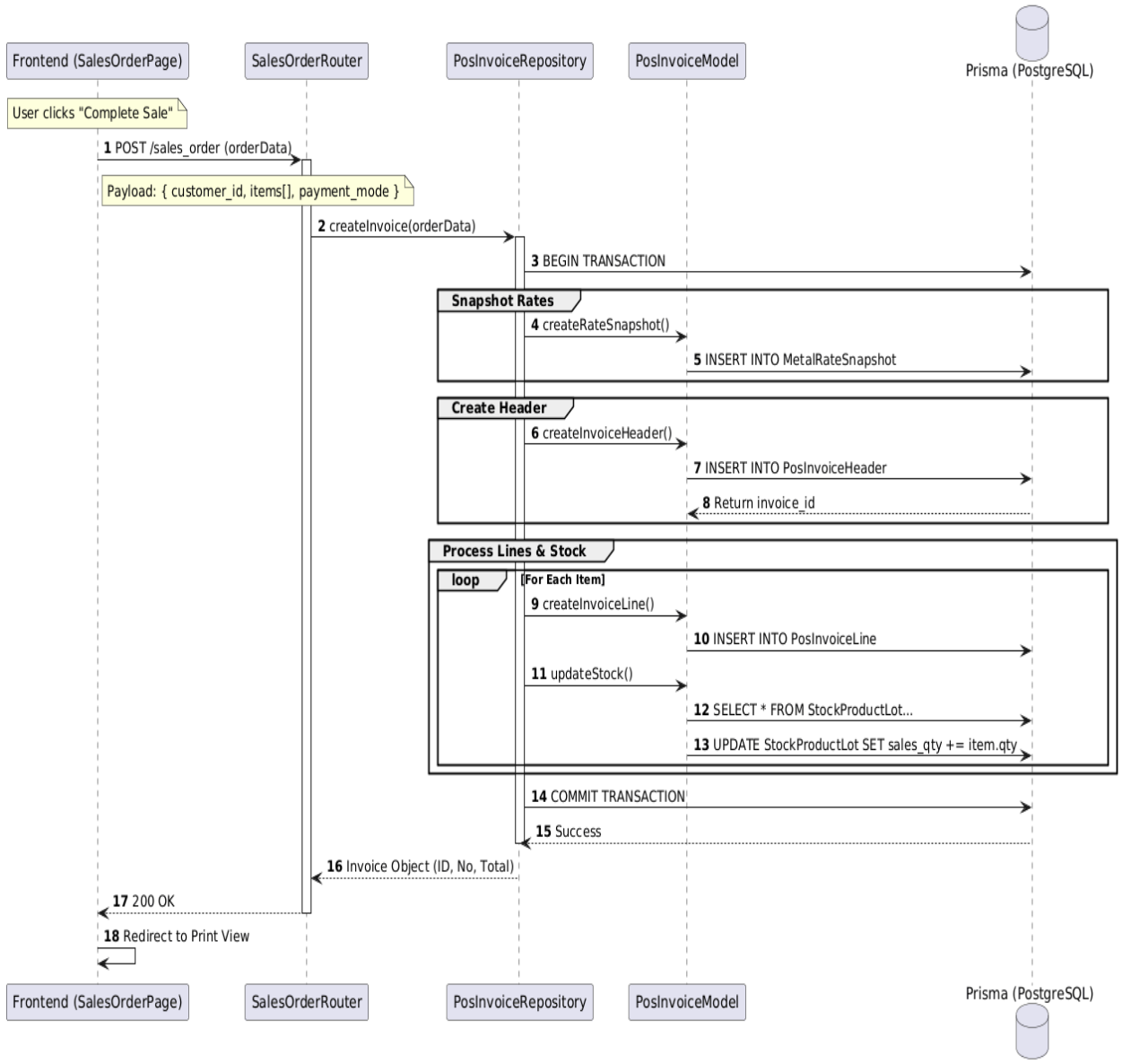


Activity 2: POS Invoice Creation (Detailed)

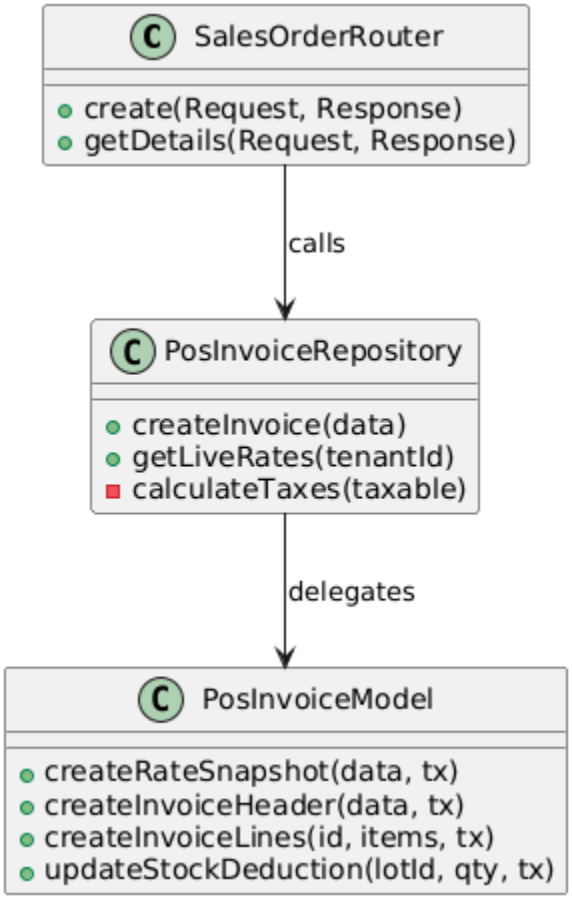


4.1.3 Sequence Diagrams

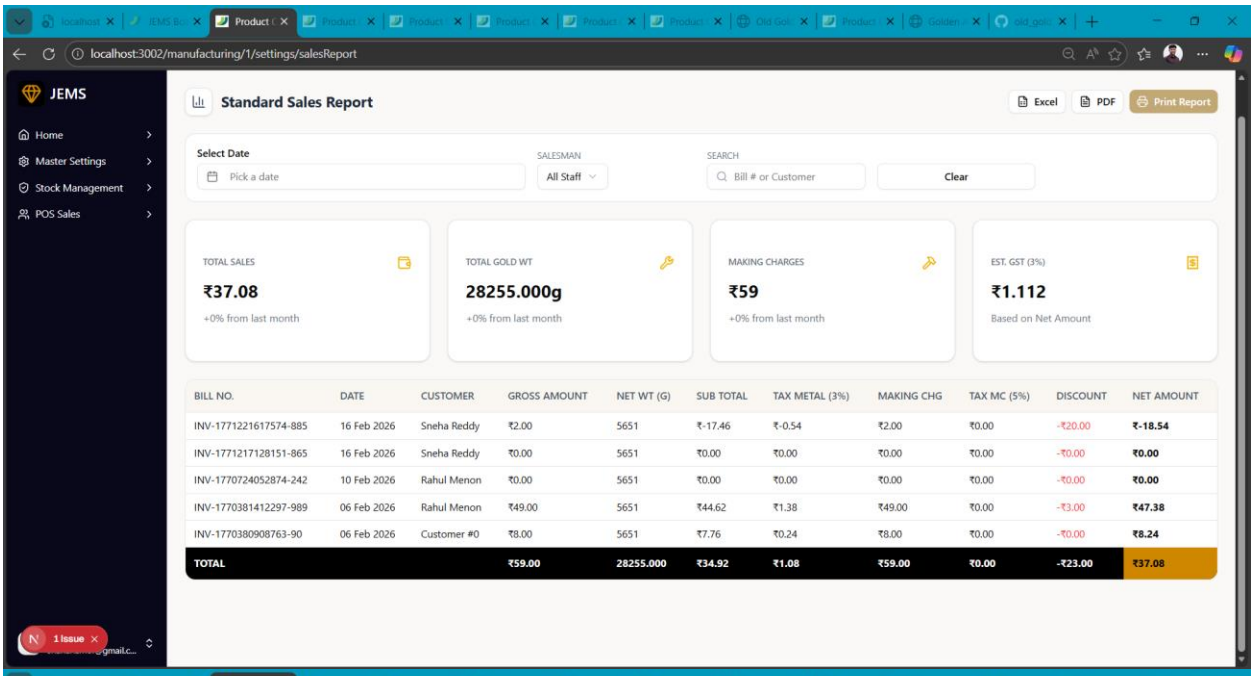
Sequence: Create Sales Order

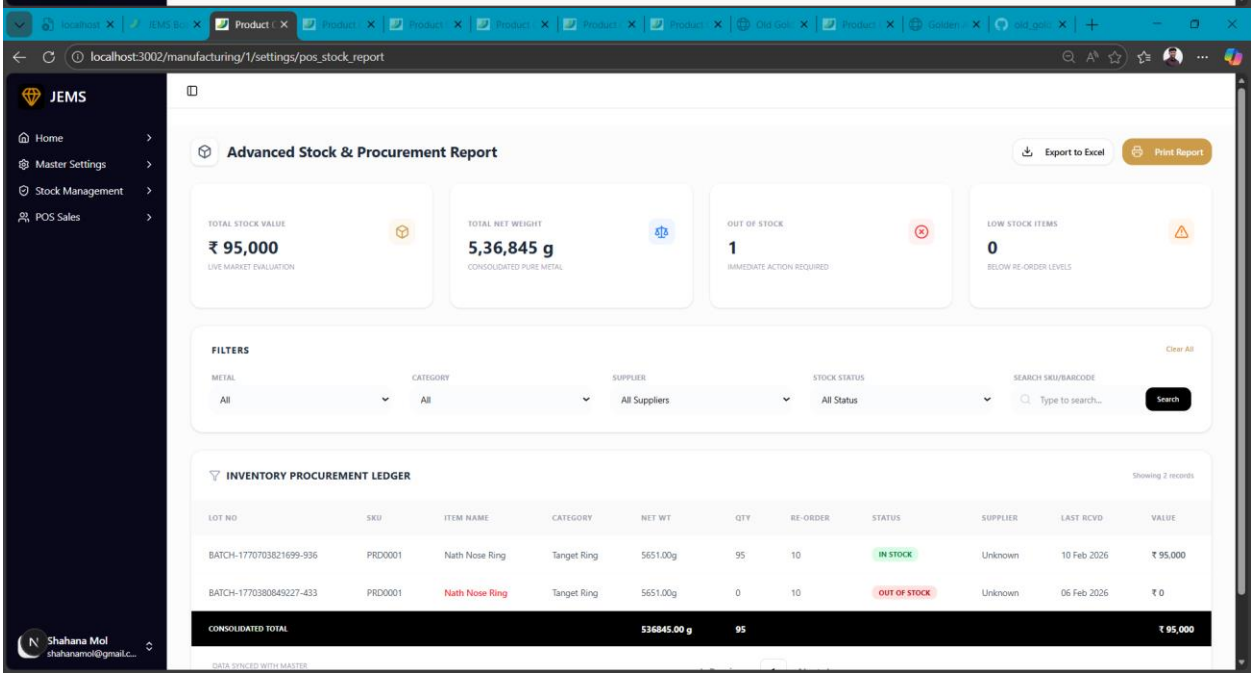


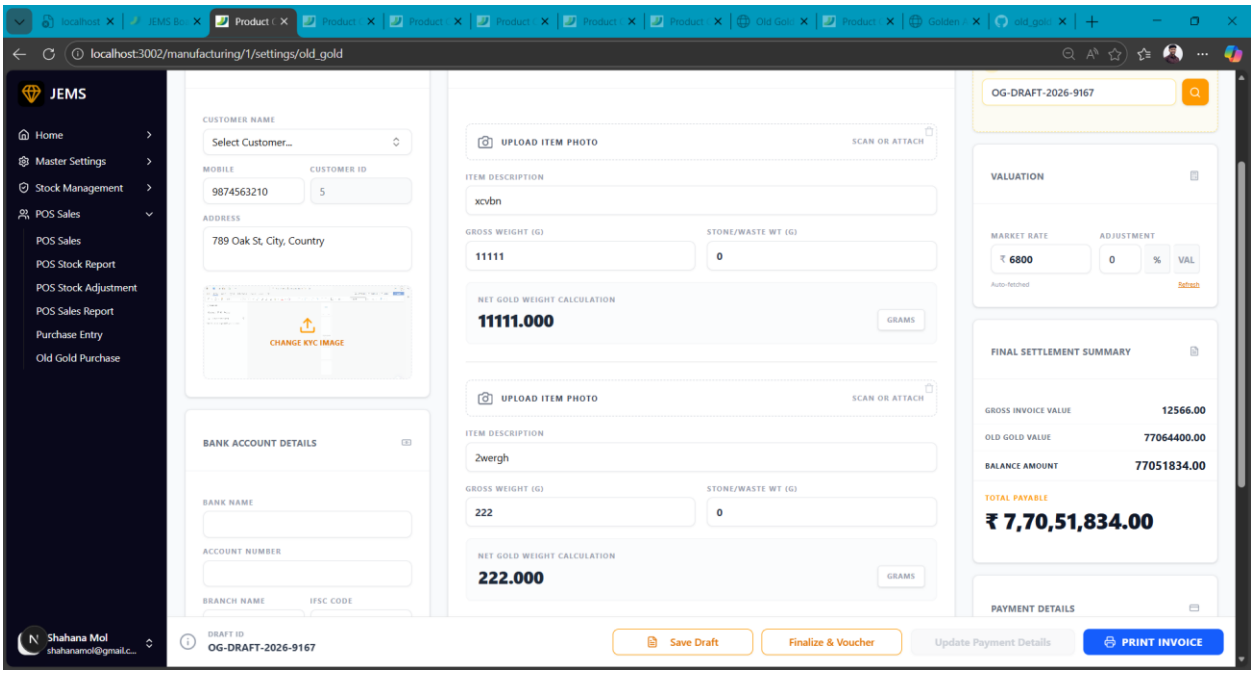
4.1.4 Class Diagrams (Design Level)


SalesOrderRouter
create(Request, Response)
getDetails(Request, Response)
calls
PosInvoiceRepository
createInvoice(data)
getLiveRates(tenantId)
calculateTaxes(taxable)
delegates
PosInvoiceModel
createRateSnapshot(data, tx)
createInvoiceHeader(data, tx)
createInvoiceLines(id, items, tx)
updateStockDeduction(lotId, qty, tx)

4.2 UI Design


localhost:3002/manufacturing/1/settings/salesReport
JEMS
Home
Master Settings
Stock Management
POS Sales
Standard Sales Report
Excel
PDF
Print Report
Select Date
Pick a date
SALESMAN
All Staff
SEARCH
Bill # or Customer
Clear
TOTAL SALES
₹37.08
+0% from last month
TOTAL GOLD WT
28255.000g
+0% from last month
MAKING CHARGES
₹59
+0% from last month
EST. GST (3%)
₹1.112
Based on Net Amount
BILL NO. DATE CUSTOMER GROSS AMOUNT NET WT (G) SUB TOTAL TAX METAL (3%) MAKING CHG TAX MC (5%) DISCOUNT NET AMOUNT
INV-1771221617574-885 16 Feb 2026 Sneha Reddy ₹2.00 5651 ₹-17.46 ₹-0.54 ₹2.00 ₹0.00 -₹20.00 ₹-18.54
INV-1771217128151-865 16 Feb 2026 Sneha Reddy ₹0.00 5651 ₹0.00 ₹0.00 ₹0.00 ₹0.00 -₹0.00 ₹0.00
INV-1770724052874-242 10 Feb 2026 Rahul Menon ₹0.00 5651 ₹0.00 ₹0.00 ₹0.00 ₹0.00 -₹0.00 ₹0.00
INV-1770381412297-989 06 Feb 2026 Rahul Menon ₹49.00 5651 ₹44.62 ₹1.38 ₹49.00 ₹0.00 -₹3.00 ₹47.38
INV-1770380908763-90 06 Feb 2026 Customer #0 ₹8.00 5651 ₹7.76 ₹0.24 ₹8.00 ₹0.00 -₹0.00 ₹8.24
TOTAL ₹59.00 28255.000 ₹34.92 ₹1.08 ₹59.00 ₹0.00 -₹23.00 ₹37.08
1 Issue


localhost:3002/manufacturing/1/settings/pos_stock_report
JEMS
Home
Master Settings
Stock Management
POS Sales
Advanced Stock & Procurement Report
Export to Excel
Print Report
TOTAL STOCK VALUE
₹ 95,000
LIVE MARKET EVALUATION
TOTAL NET WEIGHT
5,36,845 g
CONSOLIDATED PURE METAL
OUT OF STOCK
1
IMMEDIATE ACTION REQUIRED
LOW STOCK ITEMS
0
BELOW RE-ORDER LEVELS
FILTERS
Clear All
METAL All
CATEGORY All
SUPPLIER All Suppliers
STOCK STATUS All Status
SEARCH SKU/BARCODE
Type to search...
Search
INVENTORY PROCUREMENT LEDGER
Showing 2 records
LOT NO SKU ITEM NAME CATEGORY NET WT QTY RE-ORDER STATUS SUPPLIER LAST RCVD VALUE
BATCH-1770703821699-936 PRD0001 Nath Nose Ring Tanget Ring 5651.00g 95 10 IN STOCK Unknown 10 Feb 2026 ₹ 95,000
BATCH-1770380849227-433 PRD0001 Nath Nose Ring Tanget Ring 5651.00g 0 10 OUT OF STOCK Unknown 06 Feb 2026 ₹ 0
CONSOLIDATED TOTAL 536845.00 g 95 ₹ 95,000
Shahana Mol
shahanamol@gmail.c...



4.3 Theoretical Background

Technology Stack:

Frontend: Next.js (React Framework) with TailwindCSS for styling.

Backend: Node.js (Express/Next API Routes) with TypeScript.

Database: PostgreSQL with Prisma ORM.

Key Algorithms:

Weighted Average Cost: Used for inventory valuation.

FIFO (First-In-First-Out): Used for stock deduction in sales.

Zod Validation: Used for runtime schema validation of API payloads.

4.4 Database Design

Detailed Schema Catalog (Partial List of Key Tables).

Table: StockLot

Field Name	Type	Constraint	Description

Field Name	Type	Constraint	Description
lot_id	Int	PK, AutoInc	Unique Identifier
lot_no	Varchar(50)	Not Null	Barcode/Lot Number
quantity_available_weight	Decimal(12,3)		Weight currently available
raw_material_id	Int	FK	Link to Raw Material
diamond_master_id	Int	FK	Link to Diamond (if applicable)

Table: ProductMaster

Field Name	Type	Constraint	Description
product_id	Int	PK, AutoInc	Unique Identifier
design_number	Varchar(100)	Not Null	SKU Code
net_weight	Decimal(10,3)		Gold Weight
category_id	Int	FK	Link to Category

Table: PosInvoiceHeader

Field Name	Type	Constraint	Description
invoice_id	Int	PK, AutoInc	Unique Identifier
invoice_no	Varchar(50)	Not Null	Tax Invoice Number
total_value	Decimal(12,2)		Grand Total (Inc Tax)
customer_id	Int	FK	Linked Customer

Table: PosInvoiceLine

Field Name	Type	Constraint	Description
line_id	Int	PK, AutoInc	Unique Identifier
invoice_id	Int	FK	Link to Header
product_name	Varchar		Item Description

Field Name	Type	Constraint	Description
line_total	Decimal		Final Amount for Line

5. Annexure

5.1 Mathematical Formulas & Algorithms

Old Gold Valuation

$$Valuation = (GrossWeight - StoneWeight - DirtDeduction) \times (Purity\%) \times MarketRate$$

Sales Order Tax (GST)

$$TaxableValue = \sum(LineTotal_{PreTax}) - GlobalDiscount$$

$$CGST = TaxableValue \times 1.5\%$$

$$SGST = TaxableValue \times 1.5\%$$

$$GrandTotal = TaxableValue + CGST + SGST$$

5.2 Glossary / Abbreviations

Abbreviation	Explanation
SDS	Software Design Specification
POS	Point of Sale

Abbreviation	Explanation
VA	Value Addition (Making Charges)
HUID	Hallmark Unique Identification
CGST	Central Goods and Services Tax
SGST	State Goods and Services Tax

5.3 List of Figures

- Figure 1: Use Case Diagram - System Wide
- Figure 2: Activity Diagram - POS Invoice Creation
- Figure 3: Sequence Diagram - Create Sales Order

5.4 List of Tables

- Table 1: StockLot Schema
- Table 2: ProductMaster Schema
- Table 3: PosInvoiceHeader Schema

Footer Info: * Bottom Left: UCC- MCA * Bottom Right: [Page Number]